Algorithmic Trading Models, Aversion, and the Indian Scenario.

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Abstract

Algorithms are automated systems that are designed to do specific tasks. Algorithms are used in a variety of commercial domains due to their efficiency and expanding demand due to the availability of vast amounts of data. Algorithm trading, often known as algo trading, is a significant application in finance. There has been very little research on the many sorts of algorithms utilized for specialized stock trading purposes. This exploratory study examines the literature over the last five years (2019-2023) on various models of algorithms used in stock trading. The study discovered that algorithm trading models applied mostly to predicting and, to a lesser extent, classification. The most used models are Long Short-Term Memory (LSTM), Support Vector Machines (SVM), Gradient Boosting (GB), and Recurrent Neural Networks (RNN). Furthermore, algorithms have been discovered to improve the accuracy of portfolio strategies. The paper then delves into the causes of algorithm aversion and its dimensions of stock trading. The Indian environment appears to be shifting in favor of algo trading, with regulatory action to protect investors and the stock market's interests.

Keywords: algorithm trading, algo trading, algorithm aversion, algorithm in finance.

1. Introduction

'Algorithm' as a term originated from 'algorism' meant a system of numerals in Arabic' (Voiland, 2017). The origin of the term dates to middle English which refers to a Persian mathematician named "al-Khwarizmi". He contributed extensively to the fields of algebra and number systems during the ninth century. After the gradual development of al-Khwarizmi's contribution over several years, 'Alan Turing' who is called as 'Father of computer science and artificial intelligence', was the first to formally apply algorithms in computing through his invention named, 'Turing Machine' which is regarded as a model of general-purpose computers (Aron, 2024). Since then, algorithms evolved extensively to become a field generally identified

 ISBN: 978-93-83302-69-7

as "Machine Learning (ML)". ML are algorithms representing computational models that enable the system to identify patterns and predict future outcomes based on past data. This process does not demand any explicit intervention which has become the efficiency of algorithms deployed in vast business processes (Sarker, 2021). These algorithms act as an important element of artificial intelligence and perform tasks like recommendations, detection of frauds, speech recognition, picture recognition, language processing, and automated machines.

Further to the application of algorithms in various fields of business, Finance as a domain extensively applies algorithms in most of the essential processes. Financial institutions and Banks use algorithm-driven processes and models to automate customer and investor relationships, asset liability management, loan pricing, credit risk evaluation, trading, and investing processes, among other fields (Cheung, 2020).

In this paper, we focus on the application of algorithms in the trading and investment process. Algorithm trading or simply 'algo trading' refers to the execution of trade through automation by algorithms. Algo trading sets orders instantaneously, executes trades more effectively without the intervention of investor emotions, and thereby results in reduced trading expense. Due to recent developments in technology-based trading and the rapid availability of data, the need for faster execution of trades and in big volumes has grown significantly which drives more algo trading. However, there are different algorithms used in algo trading for slightly different purposes (Cohen, 2022). The need to understand the different algorithms models has grown in recent years due to its massive application in stock markets. This paper focuses on understanding different algorithms available in the literature that are used for various aspects of stock trading as well as investing.

However, there is aversion among investors to using algorithms for trading and investment purposes (Germann & Merkle, 2022). Further, post-demonetization and the COVID-19 pandemic, digitization in the financial system has rapidly grown in India. Likewise, the percentage of investors in the stock markets driven through these FinTech platforms is growing vastly. Hence, understanding the algo trading application in the Indian stock market scenario is a necessity for investors, policymakers, financial institutions, investment companies, and researchers to get clarity on the type of algorithms used and their efficiency in algo trading (Lee & Schu, 2022). This paper addresses the following research questions.,

RQ1 – What are the emerging algorithmic trading models, and their potential uses in trading and investing?

RQ2 – What are the drivers of reluctance towards algorithmic trading (algorithm aversion)?

RQ3 – What is the current condition of algorithmic trading in India, including regulatory dimensions?

The remaining parts are the paper are presented as follows. Section 2 presents the research methodology used in the study. Section 3 discusses the results of the study focusing on algo trading models used in existing literature, algorithm aversion, and the Indian scenario. Section 4 concludes with implications and scope for further research and action.

 ISBN: 978-93-83302-69-7

2. Research Methodology

This study is exploratory research in nature which is a lot qualitative and primary. This focuses on generating premises rather than testing them. It broadly acts as a foundation for further investigation into the topic. Further, existing literature like Massei (2023) and Zulkifli et al. (2023) are notable works in this regard. However, they do not exactly reflect the intended objectives of this study. For the study, the EBSCO database and Google Scholar are used to identify suitable literature. Likewise, the study used the PRISMA approach to collect the relevant literature for further investigation. PRISMA stands for Preferred Reporting Items for Systematic Reviews and Meta-Analyses. The PRISMA statement's overarching purpose is to improve the transparency and scientific validity of a reported systematic review or meta-analysis (Swartz, 2011). The literature is collected during the period of the last five years (2019 -2023) since the previous studies have more focused on works from previous periods. The process of literature collection is depicted in Figure 1.

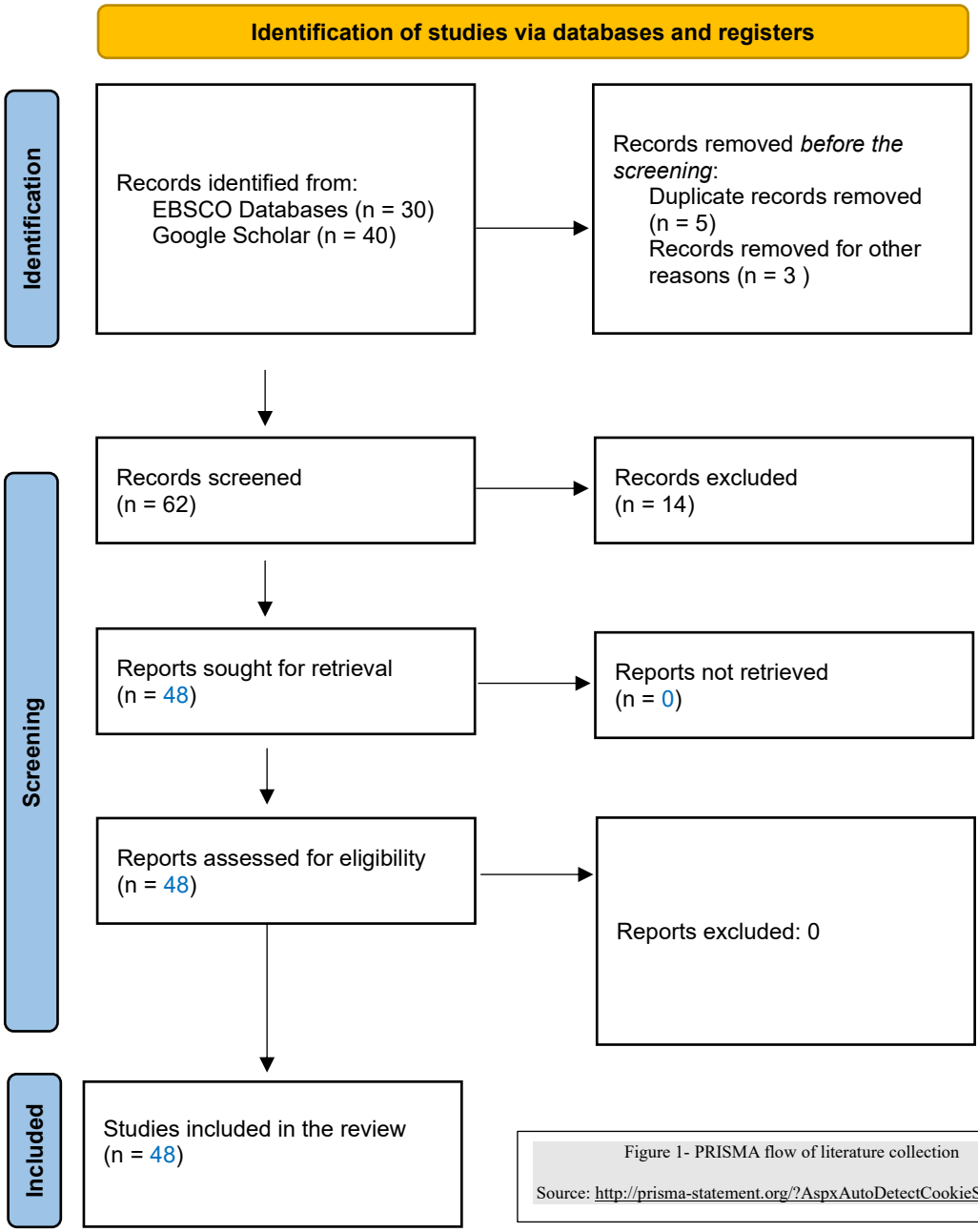

Figure 1- PRISMA flow of literature collection

Source: http://prisma-statement.org/?AspxAutoDetectCookieSupport=1

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3. Results and Discussion

On examining the collected literature, the discussion section is divided into the application of algorithms in trading, and the application of algorithms in investment. The primary aim in trading is stock price prediction whereas in investment it is about managing portfolios. Followed by the literature evidence of algorithm aversion and the Indian scenario are discussed.

3.1 Application of algorithms –

3.1.1 Stock Trading

Algorithms play a significant role in stock market trading mostly on stock prediction. The stock market data are time series in nature that are noisier, and non-stationary. Due to this nature, stock market forecasting has become a more challenging topic of study. The classic financial theory named Efficient Market Hypotheses (EMH) says that outperforming the stock markets consistently for a longer period is not possible. However, recent technological developments have drastically improved the efficiency of traders to outperform the markets largely (Baboc & Dinica, 2013).

Due to this fact, researchers and investors are interested in understanding the interrelated components of the markets more efficiently. However, a stock market is widely impacted by various macroeconomic factors such as economic growth, forex, political changes, policy implications, foreign investors, fund flows, investor behavior, etc. Just being aware of these elements and guessing how they influence the stock market, however, is insufficient.

Such elements must be integrated into a model for forecasting by transforming the provided data into meaningful characteristics, which are vital in algorithm-based approaches. These have contributed significantly to the discipline of time series financial prediction. A stock market time series is a suitable yardstick for measuring how well such algorithm predictions perform because of their complexity and fluctuations. Algorithms have been applied to forecast various aspects such as stock price, stock return, and fluctuations or volatility (Sonkavde et al., 2023). Likewise, they have been applied not just in forecasting individual stocks but in indices as well. Further, studies have applied algorithms in the classification of stock price movements as binary categorizations such as uptrend or downtrend, bullish or bearish, etc.

3.1.2 Stock forecasting

The research work of Sun, Xiao, Liu, Zhou, and Xiong (2019) focused on stock price prediction instantaneously. Many of the studies have applied algorithms to predict returns from a stock or index the next day. Likewise, Khattak, Ali, and Rizvi, (2021) applied algorithms to predict the impact of various predictors of the stock market. In a similar study, Zaremba, Kizys, Tzouvanas, Aharon, and Demir (2021) investigated the impact of the COVID-19 pandemic. Notable studies broadly focused on creating support for mobile applications that notify potential investors of potential market close prices and creating trading strategies.

 ISBN: 978-93-83302-69-7

Following are the notable research works focused remarkably on stock forecasting. Discussion on the algorithm models and the context of the studies are presented precisely. Zhu, Zhang, Liu, Wu, and Wang (2019) used a Bidirectional gated recurrent unit (BiGRU) to investigate the Shanghai Futures Exchange (SHFE) in China. BiGRU is an orderly processing framework made up of two GRUs. One takes input in one way and the other in the other direction. Only the input and forget gates are present in this bidirectional recurrent neural network (Paperswithcode, 2024). In a similar study, Maqsood et al. (2020) employed Linear Regression (LR), Support Vector Machine (SVM), and Deep Learning (DL) to anticipate equities and indices in the United States, Hong Kong, Turkey, and Pakistan. LR is a linear technique for modeling the relationship between a scalar response to and one or more explanatory factors. SVM is a strong machine-learning technique that could be used for nonlinear or linear classification, regression, and even outlier detection (Kanade, 2022). Deep Learning (DL) refers to the usage of many network layers in neural network architecture.

In a notable work, Nevasalmi (2020) predicted the S&P 500 index using the K-NN classifier, Gradient Boosting (GB), Random Forest (RF), Neural Network (ANN), and SVM. K-NN classifier refers to the k-nearest neighbor algorithm, which is a non-parametric, supervised learning classifier that employs distance to classify or predict the clustering of a single data point. GB is one of the most popular forward-learning ensemble methods used in prediction algorithms. Random forests are a method of averaging numerous deep decision trees that have been trained on different regions of the same training set to reduce variation (Brownee, 2020).

Kamdem, Essomba, and Berinyu (2020) predict the commodity price movements of crude oil, wheat, and silver using the LSTM model. LSTM refers to Long Short-Term Memory that, unlike standard neural networks, has feedback connections, enabling it to interpret complete data sequences rather than each point of data. As a result, it is extremely successful in comprehending and foreseeing patterns in sequenced information such as time-series data.

Bhandari et al. (2022) forecast the S&P 500 and Dow Jones Industrial Average, of the US using Upgraded LSTM. Furthermore, Zhang, Chu, and Shen (2021) used the LSTM model to computerize the trading of 103 stocks on China's SSE index. Likewise, Ayala, Garcia-Torres, Noguera, Gomez-Vela, and Divina (2021) employed Multivariate Linear Regression, Neural Networks, RF, and Support Vector Regression (SVR) to automate trading of indexes from different regions, such as the Ibex35 (IBEX) in Spain, the DAX in Germany, and the Dow Jones Industrial (DJI) in the United States.

Keyan, Jianan, and Dayong (2021) forecasted 46 equities from the SSE 50 in China using RNN (Recurrent Neural Networks), LSTM, and GRU. Khattak, Ali, and Rizvi (2021) forecasted the EURO STOXX 50 index in Europe using LASSO. LASSO stands for Least Absolute Shrinkage and Selection Operator which is a regression analysis method that uses choosing variables and normalization to improve the predictability and interpretability of the produced quantitative model. Gradient Boosting Regression, GLM (Generalized linear models), and Deep Learning were utilized by Ingle and Deshmukh (2021) in stock price forecasting. Similarly, Zhang and Lou (2021) forecasted small and medium-sized boards in China using BPNN (Back Propagation Neural Network). BPNN is a multilayer feed-forward neural network that is trained according to an error backpropagation algorithm. Backpropagation is the neural

 ISBN: 978-93-83302-69-7

network training process of feeding error rates back through a neural network to make it more accurate. Akhtar, Zamani, Khan, Shatat, and Dilshad (2022) anticipate global indices such as the BSE 500, DJIA, NASDAQ, India, and the US using RF, SVM, and LSTM.

Gupta, Bhattacharjee, and Bishnu (2022) forecasted the CNX Nifty in India using the GRU-based model. Ma and Yan (2022) forecasted the Shanghai stock index in China using CNN. CNN refers to a convolutional neural network which is a regularized sort of feed-forward neural network that masters attributes via optimizing filters. The use of normalized weights over fewer connections prevents disappearing and inflating gradients, which were observed during backpropagation in normal neural network models. In a notable work, Bhandari et al. (2022) used LSTM to calculate the S&P 500 index in the United States.

Text Mining Techniques - Text mining is becoming more prevalent in the big data business because of its ability to retrieve vast volumes of pertinent data from speeches, posts, information, and other social media platforms that can enhance stock prediction (Hassani et al. 2020). In a similar vein, the goal of market sentiment as a metric is to portray the common attitude of people on a specific scenario published on social media sites about an individual stock or the overall market. It primarily requires determining the opposing views of statements—that is, whether investors have a positive, adverse, or indifferent opinion of the stock market—based on a social media release.

Maqsood et al. (2020) evaluated the effects of important worldwide and local event sentiments on several stocks from the United States, Hong Kong, Turkey, and Pakistan using a Twitter dataset. The results they found demonstrate how the economic and political landscape of the country, as well as public opinions, influence the volatility of the stock market. Local events have a higher impact on forecasting algorithmic efficiency than global occurrences.

Saurabh and Dey (2020) use ANN to determine the association among the social sentiments of individuals on the FTSE 100 index. They point out that the "contented" element could significantly improve the stock market index return prediction.

Keyan, Jianan, and Dayong (2021) propose a further function that uses user information as well as the shares they monitor on a private trading platform to establish an "investor-stock" relationship that develops deep learning models. Each of the three models i.e., Long Short-Term Memory (LSTM), Recurrent Neural Networks (RNN), and Gradient Recurrent Unit (GRU) exhibits improved accuracy when the network variable is integrated into conventional variables rather than just including conventional variables. By providing two distinct investor attention substitutes, Zhang, Chu, and Shen (2021) show a lack of important features. Along with presenting ongoing trading, funds, and traditional features, they include media coverage, or the number of daily news pieces released on a certain stock, as well as anomalous search activity. When these features were incorporated into the Long Short-Term Memory (LSTM) model, they enhanced the precision of the closing estimate for the SSE 50 stocks.

Baek, Mohanty, and Glambosky (2020) explore the impact of economic factors on US volatility in equity markets using a Decision Tree, a Genetic Algorithm, and a Random Forest (RF) method. Random Forest is used by Lohrmann and Luukka (2019) to choose characteristics in intraday return on investment trend forecasting.

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3.1.3 Stock Investing

As Investing we refer to the aspects of investment portfolio management. Portfolio management comprises a wide range of duties, including asset selection, asset prioritization based on tolerance for risk and anticipated returns, and the development of suitable portfolio methodologies. It is also necessary to adjust these approaches and reset portfolio composition regularly to achieve long-term or short-term financial objectives. Because of the numerous responsibilities, the phrases choosing a portfolio, portfolio distribution, and optimizing a portfolio have been used alternately in earlier studies. However, they all refer to the same fundamental element of managing a portfolio (Ozbayoglu, Gudelek, & Sezer, 2020).

Methods from machine learning are used in Mean-Variance investment design by Paiva, Cardoso, Hanaoka, and Duarte (2019), Chen, Zhang, Mehlawat, and Jia (2021), and Wang, Li, Zhang, and Liu (2020). Picasso, Merello, Ma, Oneto, and Cambria (2019) used technical evaluation along with opinions from headlines to predict the behavior of 20 corporations on the NASDAQ100 index. They could achieve annualized returns of more than 80% by applying trade models. The study advances the combination of both fundamental and technical evaluation and facilitates the creation of new trading methods.

Barua and Sharma (2022) build strategies based on historical data from eleven MSCI Asia Pacific sector indexes obtained from the Bloomberg dataset. (Betancourt & Chen, 2021) investigated Binance cryptocurrency data, which includes minute, hourly, and daily prices for numerous cryptocurrencies. In contrast, Vo, He, Liu, and Xu (2019) rebuilt the algorithms after each evaluation period, i.e., monthly, and annually. Chen, Zhang, Mehlawat, and Jia (2021) utilized this concept to build a portfolio of 6 securities, while Wang, Li, Zhang, and Liu (2020) used ten stocks.

Tan, Yan, and Zhu (2019) selected 20 firms to establish an investing portfolio to be retained for 20 days, and Vo, He, Liu, and Xu (2019) experimentally tested that an ESG portfolio of 12 equities performed well. Random Forest (RF) outperforms Support Vector Machine (SVM), Long short-term memory (LSTM), Convolutional Neural Network (CNN), Multi-Layer Perceptron (MLP), and Autoregressive integrated moving average (ARIMA) approaches in the management of portfolio strategy, according to the latest research (Nazareth and Reddy, 2023).

3.2 Algorithm Aversion -

The application of algorithms has grown rapidly in recent years. However, there are issues concerning its acceptance in various fields. This is very evident in the fields of finance and especially in trading and investments as well.

Algorithm aversion, defined by Jussupow, Benbasat, and Heinzi (2020), is a "prejudiced evaluation of an algorithm reflected in unfavorable behaviors and attitudes against the algorithm in contrast to a human agent." The tendency to disregard algorithms has been documented in a variety of domains. For instance, in medical treatment, physician advice is

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rated higher than computer-assisted or physician-assisted by computer (algorithm) recommendations.

Logg, Minson, and Moore (2019) have identified algorithm satisfaction while executing non-financial tasks. According to Castelo, Bos, and Lehmann (2019), algorithm aversion is stronger for subjective actions (such as delivering humor) than for objective activities (such as buying stocks). When dealing with logical problems, users almost always use algorithms rather than people.

According to the research, the alternative to using an algorithm is to use a personal expert or oneself. Overconfidence and a sense of control may lessen reliance on an algorithm in the long run. In several studies, humans and algorithms are classified and characterized differently. Making major references to their ability or reasoning may influence the choice. Also, perplexing is task subjectivity and whether events are examined before or after feedback on performance. In an experimental study,

According to Germann and Merkle (2022), financial service providers see algorithms as a supplement to human investment managers rather than a rival. These remarks imply that human expertise is regarded to benefit Robo-advisors or rule-based funds. Financial service providers could use a hybrid method in which human specialists supervise algorithms and act in severe cases.

3.3 The Indian scenario –

Algorithmic trading began appearing in India about 2010 and at first, was limited to institutions and brokers. However, with the rise of digital discounters the retail business has gained free access to constructing algorithms, and the possibilities are limitless (Anand, 2022). The National Stock Exchange of India (NSE) and the Bombay Stock Exchange (BSE) are two exchanges and brokers in India that provide algo-trading platforms and services. Omnesys Nest, NOW, and Tradeplus are three famous algo-trading platforms in India (Live mint, 2022).

The possibility of errors or faults in the algorithms or computer systems used to execute trades could result in investor losses. Algo trading may also be prone to market manipulation and other sorts of fraud. In recent years, India recently learned about the infamous NSE Scam (Joshi, 2022). SEBI, the stock market regulator in India, the primary issue with algo trading appears to be mis-selling of algo services and strategies, particularly when it comes to quoting past performance, forecasting future performance, and assigning ratings (Economic Times, 2022). Experts opine that rather than a practical prohibition on algo trading, the regulator (SEBI) must order that all performance indicators be accompanied by proper risk disclosures to limit the risks.

4. Conclusion

Raising data availability and technological advancement largely drive the application of algorithm models for the prediction and automation of various business tasks. Likewise, in the

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fields of finance algorithms are applied in process automation, fraud detection, assets liability management, and most importantly prediction of stock movements. Algo trading broadly specifies the application of algorithms in trading-related aspects like forecasting share prices and trends (Cohen, 2022). However, there are fewer comprehensive works available to understand the algorithm models applied for trading.

Thus, this study examined the most recent literature to investigate the various algorithm models applied in different contexts with respective tasks. From the literature evidence, it is found that Long Short-term Memory (LSTM), Gated recurrent unit (GRU), Support Vector Machine (SVM), Gradient Boosting (GB), Random Forest (RF), and Backpropagation Neural Network (BPNN) are the algorithm models mostly applied in algo trading purpose (Maqsood et al., 2020; Nevasalmi, 2020; Brownee, 2020; Kamdem, Essomba, & Berinyu, 2020; Bhandari et al. 2022). Further, text analytics aspects like sentiment analysis on social media textual content were used in the literature to improve the accuracy of the prediction of stock prices and scenarios. Further to direct stock price forecasting, investing processes like portfolio management have been studied by researchers to improve the efficiency of implementing portfolio strategies.

However, the practice of depending on algorithm-based decisions or automation is not agreeable to all. The literature evidence states that aversion to algorithms in stock trading reflects the fact that investors must follow hybrid strategies of algorithmic suggestion driven by human intervention (Germann & Merkle, 2022). The study briefly addressed the present scenario of algo trading in the Indian context. The use of algorithms is emerging in Indian markets supported and monitored by the regulatory environment. However, not so many research works have been identified to reflect the application and acceptance of algorithms in stock market trading in India. More work of research must be empirically focused on uncovering the application of specific models of algorithms, accuracy, their acceptance, and associated risks in the Indian stock market environment.

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